Alcohols [-OH]

* classification according To number of hydroxyl group

monohydroxyl alcohols
CH_2-CH_2-OH ethanol
CH_3-OH methanol

Dihydroxyl alcohols
CH_2-OH
CH_2-OH
ethylene glycol

Polyhydroxyl alcohols
CH_2-OH
$CH-OH$
CH_2-OH
glycerol

classification according to Number of hydrogen attached to α carbon

Primary $R-CH_2-OH$
Secondary $R-CH-OH$
Tertiary $R-C-OH$

1) oxidation test.

transfer of electrons

$3\,CH_3-CH_2-OH + K_2Cr_2O_7 + 4H_2SO_4 \longrightarrow 3\,CH_3-CHO + K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O$

$3\,CH_3-OH + 2K_2Cr_2O_7 + 8H_2SO_4 \longrightarrow 3\,H-COOH + 2K_2SO_4 + 2Cr_2(SO_4)_3 + 11\,H_2O$

$R-OH \xrightarrow{O_2} R-CHO \xrightarrow{O_2} R-COOH$

→ 1ml of unknown alcohol
→ 1ml of dil H_2SO_4
→ 1ml of $K_2Cr_2O_7$

heating in waterbath for (5-10) minute

yellow color turns into green → Reduction of $Cr^{+6} \rightarrow Cr^{+3}$

Rotax test

Pink Red → colorless

(1) → 5 drops of Phenol Phthalene Ph.Ph
(2) → 5 drops of
(3) → 1ml of Borax
(4) heating till → colorless → Pink red

→ Iodoform test:

$CH_3-CH_2OH + I_2 \xrightarrow{NaOH} CH_3-CHO$ (استيك)

$CH_3-CHO \xrightarrow{I_2/A} CI_3-CHO$

$CI_3-CHO \xrightarrow{NaOH/A} CHI_3 + HCOONa$ (hydrolysis)

→ drops of I_2 + NaOH sol → Brown → Yellow color
→ 5ml of I_2
→ 1ml of alcohol heating in water bath [5-10] Yellow color

* Esterification Test

1) ethanol

$CH_3-CH_2OH + CH_3-COOH \xrightarrow{conc.\ H_2SO_4} CH_3-COOCH_2-CH_3 + H_2O$
ethyl acetate

→ 1ml of conc H_2SO_4
→ 1ml of acetic
→ 1ml of ethanol
heating in water bath for [5-10] minute → dr. fl → 5ml of $NaHCO_3$ effervescence and the solution of a bad apple odour → ethanol →

2) methanol

$CH_3-OH + C_6H_4(COOH)(OH) \xrightarrow{conc\ H_2SO_4} C_6H_4(COOCH_3)(OH) + H_2O$
methyl salicylate

→ 1ml conc H_2SO_4
→ ½ gm of Salicylic acid → heating in water bath [5-10] minute → Effervescence and evolution of oil winter green → 5ml of $NaHCO_3$
→ 1ml of methanol

3.

Pink Red → colorless)

drops of Phenol Phthalene Ph.Ph

1ml of Borax

heating till → colorless → Pink red

→ Iodoform test

$CH_3-CH_2OH + I_2 \xrightarrow{NaOH} CH_3-CHO$ (اسيتالدهيد)

$CH_3-CHO \xrightarrow{I_2/OH^-} CI_3-CHO$

$CI_3-CHO \xrightarrow{NaOH} CHI_3 + HCOONa$

drops of NaOH-Iodi → Brown → Yellow color

5ml of I_2

1ml of alcohol heating in water bath [5-10]

Yellow color

* Esterification Test

1) ethanol

$CH_3-CH_2-OH + CH_3-COOH \xrightarrow{Conc\ H_2SO_4} CH_3-COOCH_2-CH_3 + H_2O$

ethyl acetate

1ml of conc H_2SO_4

1ml of acetic

1ml of ethanol

heating in water bath for [5-10] minute

drops → 5ml of $NaHCO_3$

effervescence and solution of a bad apple odour → ethanol →

2) methanol

$CH_3-OH + C_6H_4(COOH)(OH) \xrightarrow{Conc\ H_2SO_4} C_6H_4(COOCH_3)(OH) + H_2O$

methyl salicylate

1ml Conc H_2SO_4

½ gm of Salicylic acid → heating in water bath [5-10] minute →

1ml of methanol

Effervescence and solution of oil winter green

5ml of